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Case Study Report: Banking System with Simple Interest Calculator

1. Introduction to the Case Study

Banking systems are an essential part of modern financial infrastructure. They help manage customer accounts, deposits, withdrawals, and financial calculations efficiently. In programming education, implementing a simple banking system allows students to understand real-world applications of programming concepts such as object-oriented programming, file handling, and algorithmic thinking.

This case study focuses on developing a ***Banking System using C++*** that allows users to create bank accounts, deposit and withdraw money, search for accounts, and calculate simple interest. The system also stores account records in a file for future reference.

The project demonstrates how programming can be applied to financial operations and helps build practical problem-solving skills in software development.

2. Case Background (Abstract)

In traditional banking systems, managing financial records manually can be time-consuming and prone to errors. A computerized system simplifies the process by automating calculations and maintaining organized records.

The objective of this case study is to design and implement a **console-based banking application** that performs basic banking operations including:

- Creating a customer account
- Depositing money
- Withdrawing money
- Searching for an account
- Calculating simple interest
- Saving and viewing records using file handling

The system should ensure accurate calculations and provide a user-friendly interface through a menu-driven program. It also includes error handling to prevent invalid inputs and incorrect operations.

3. Case Study Design

The system is designed using **Object-Oriented Programming (OOP)** principles in C++. A class named **BankAccount** is created to store customer details and perform banking operations.

System Components

Class:
BankAccount

Attributes (Data Members):

- Customer Name
- Account Number
- Principal Amount
- Interest Rate
- Time Period
- Calculated Interest

Functions (Member Methods):

- Create Account

- Deposit Money
- Withdraw Money
- Search Account
- Calculate Simple Interest
- Display Interest Details
- Save Records to File
- View Stored Records

Program Structure

The system operates using a menu-driven interface, where users select an option to perform a specific banking operation.

Menu options include:

1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

The case study uses several programming techniques and algorithms to implement the banking system.

1. Object-Oriented Programming (OOP)

The system is built using the concept of **classes and objects**. The BankAccount class encapsulates all customer data and related operations.

2. Simple Interest Formula

The financial calculation used in this system is the **Simple Interest Formula**:

$$Simple\ Interest = Principal \times Rate \times Time / 100$$

This formula calculates the interest earned on a deposited amount over a specific time period.

3. File Handling

The program uses file handling to store customer records permanently using:

- **ofstream** for writing data
- **ifstream** for reading stored records

The records are saved in a file named: bank_records.txt

4. Conditional Statements

Conditional statements such as **if-else** are used to validate user inputs and prevent invalid transactions such as:

- Negative deposit amounts
- Invalid account numbers
- Withdrawals exceeding account balance

5. Menu Driven Control Structure

The program uses a do-while loop and switch statement to repeatedly display the menu and execute the selected operation.

5. Case Study Implementation Details and Snapshots

The implementation of the banking system was carried out using C++ programming language.

Development Environment

Programming Language: C++
Compiler: g++ / CodeBlocks / VS Code
Operating System: Windows / macOS / Linux

Implementation Steps

1. Creation of the BankAccount class
2. Implementation of account creation functionality
3. Implementation of deposit and withdrawal operations
4. Integration of simple interest calculation
5. Implementation of account search functionality
6. Implementation of file handling for record storage
7. Development of a menu-driven user interface

Sample Output Snapshot

```
irenek@Irenes-MacBook-Air CaseStudy % ./banking

======= MINI BANKING SYSTEM =======
1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit
Enter your choice: 1

Enter Customer Name: Irene
Enter Account Number: 101
Enter Deposit Amount (Principal ₹): 10000

======= MINI BANKING SYSTEM =======
1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit
Enter your choice: 2

Enter Rate of Interest (% per annum): 5
Enter Time Period (in years): 10

----- Interest Details -----
Customer Name: Irene
Account Number: 101
Principal Amount: ₹10000.00
Simple Interest: ₹5000.00
Total Amount After Interest: ₹15000.00

Record saved successfully!
```

```
======= MINI BANKING SYSTEM =======
1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit
Enter your choice: 4

Enter Account Number: 101
Enter amount to deposit: ₹5000
Deposit successful! New Balance: ₹15000.00

======= MINI BANKING SYSTEM =======
1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit
Enter your choice: 5

Enter Account Number: 101
Enter amount to withdraw: ₹3000
Withdrawal successful! Remaining Balance: ₹12000.00

======= MINI BANKING SYSTEM =======
1. Create Account
2. Calculate Simple Interest
3. View Saved Records
4. Deposit Money
5. Withdraw Money
6. Search Account
7. Exit
Enter your choice: 2

Enter Rate of Interest (% per annum): 3
Enter Time Period (in years): 2

----- Interest Details -----
Customer Name: Irene
Account Number: 101
Principal Amount: ₹12000.00
Simple Interest: ₹720.00
Total Amount After Interest: ₹12720.00

Record saved successfully!
```

CaseStudy > bank_records.txt

```
Name: Irene | Account No: 101 | Principal: ₹10000 | Rate: 5% | Time: 10 years | Interest: ₹5000 | Total: ₹15000
Name: Irene | Account No: 101 | Principal: ₹12000 | Rate: 3% | Time: 2 years | Interest: ₹720 | Total: ₹12720
Name: Caryn | Account No: 102 | Principal: ₹5000 | Rate: 5% | Time: 2 years | Interest: ₹500 | Total: ₹5500
Name: Olivia | Account No: 103 | Principal: ₹150000 | Rate: 6% | Time: 3 years | Interest: ₹27000 | Total: ₹177000
Name: Charles | Account No: 104| Principal: ₹70000 | Rate: 10% | Time: 4 years | Interest: ₹28000 | Total: ₹98000
```

6. Case Study Results and Conclusion

The mini banking system was successfully implemented using C++. The program allows users to perform basic banking operations efficiently through a menu-driven interface.

The system demonstrates the practical use of programming concepts such as object-oriented programming, file handling, and financial calculations.

Key outcomes of the project include:

- Accurate calculation of simple interest
- Successful implementation of account creation and management
- Secure handling of deposit and withdrawal transactions
- Efficient storage and retrieval of records using file handling

Overall, the project highlights how programming can be applied to solve real-world financial problems and provides a strong foundation for building more advanced banking software systems.

7. References

1. Programming Concepts taught in B.Tech CSE curriculum
2. Let Us C++ by Yashwant Kanetkar
3. C++ Documentation – www.cplusplus.com
4. Lecture Notes and Classroom Materials